
Employment

- 2025 – Pres. **Software Engineer**, *Databricks Inc.*, Mountain View, CA, United States.
Apache Spark, PySpark, and the Spark Experience team.
- 2025 – Pres. **Adjunct Assistant Professor**, *CICS at UMass Amherst*, Amherst, MA, United States.
AI Lakehouse, Lakebase, AI systems, and AI-native data engines.
- 2024 **Software Engineer Intern**, *Observe Inc.*, San Mateo, CA, United States.
Dataset Transformers, Go, Snowflake
- 2022 **Research Intern**, *Visa Research, Visa Inc.*, Palo Alto, CA, United States.
SWAT-RT: Real-time Window Aggregation on Streaming Systems, C++, Redis, Kafka, Flink
- 2020 **Research Intern**, *Infrastructure System Lab, ByteDance Inc.*, Mountain View, CA, United States.
HTAP Database System, C++, Kudu, MySQL

Education

- 2019 – 2025 **Ph.D., Computer Science**, *University of California, Irvine*, CA, United States.
Lab: [Information Systems Group \(ISG\)](#); Advisor: [Dr. Chen Li](#); GPA: 3.96/4.0.
Research Interest: Data-Processing Systems, Database Management Systems, Machine Learning Systems.
Dissertation: *UDF-Centric Dataflow Systems for Supporting User-Defined Functions in Collaborative Data Science, AI, and ML*.
- 2015 – 2019 **B.S., Computer Science**, *University of California, Irvine*, CA, United States.

Awards

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|------|---|-----------|
| 2026 | CS Best Dissertation Honorable Mention Award | UC Irvine |
| 2025 | Most Promising Future Faculty Second Place Award | UC Irvine |
| | Joseph & Dorothy Fischer Memorial Endowed Fellowship | UC Irvine |
| | Beall Family Foundation Graduate Student Entrepreneur Award in Computer Science | UC Irvine |
| 2024 | Best Demo Award Runner-Up | SIGMOD |
| | Graduate Dean's Dissertation Fellowship | UC Irvine |
| | Student Travel Award | SIGMOD |
| 2023 | Public Impact Fellowship | UC Irvine |
| | Student Travel Award | VLDB |
| 2020 | Best Lecturer Award | CUCS |

Work Under Submission

- 2026
- S3 **Peanut: Port-Based Compilation & Scheduling of UDFs in Dataflow Systems**
[Yicong Huang](#), [Jiadong Bai](#), and [Chen Li](#)
In submission to VLDB
- S2 **SWAT-RT: Low Latency Windowed Analytical Feature Enrichment on Real-time Streams with Out-of-order Support**
[Yicong Huang](#), [Mangesh Bendre](#), [Robert Christensen](#), [Mahashweta Das](#), [Chen Li](#), and [Hao Yang](#)
In submission to MLSys
- S1 **How Can AI Help Me with Data Science? An Exploration of AI-Assistants on Data Workflow Systems**
[Xi Lu*](#), [Yicong Huang*](#), and [Chen Li](#)
In submission to SIGCHI [*The first two authors share equal contributions]

Computer Science Publications

2026

D4 Click, Share, Learn: Teaching Data Science Using Apache Texera

Sarah Asad, Xiaozhen Liu, Kun Woo Park, Xinyuan Lin, Jiadong Bai, Shengquan Ni, Yicong Huang, and Chen Li
Proceedings of the International Conference on Extending Database Technology (EDBT) [to appear]

2025

W2 ML-Asset Management: Curation, Discovery, and Utilization

Mengying Wang, Moming Duan, Yicong Huang, Chen Li, Bingsheng He, and Yinghui Wu
Proc. VLDB Endow., Vol. 18, No. 12

C6 IcedTea: Efficient and Responsive Time-Travel Debugging in Dataflow Systems

Shengquan Ni, Yicong Huang, Zuozhi Wang, and Chen Li
Proc. VLDB Endow., Vol. 18, No. 3, pp. Pp. 902–914

C5 Pasta: A Cost-Based Optimizer for Generating Pipelining Schedules for Dataflow DAGs

Xiaozhen Liu, Yicong Huang, Xinyuan Lin, Avinash Kumar, Sadeem Alsudais, and Chen Li
Proc. ACM Manag. Data (SIGMOD), Vol. 2, No. 6, pp. 248:1–248:26

C4 Texera: A System for Collaborative and Interactive Data Analytics Using Workflows

Zuozhi Wang, Yicong Huang, Shengquan Ni, Avinash Kumar, Sadeem Alsudais, Xiaozhen Liu, Xinyuan Lin, Yunyan Ding, and Chen Li
Proc. VLDB Endow., Vol. 17, No. 11, pp. Pp. 3580–3588

2024

C3 Udon: Efficient Debugging of User-Defined Functions in Big Data Systems with Line-by-Line Control

Yicong Huang, Zuozhi Wang, and Chen Li
Proc. ACM SIGMOD, Vol. 1, No. 4, pp. 225:1–225:26

W1 Data Science Tasks Implemented with Scripts versus GUI-Based Workflows: The Good, the Bad, and the Ugly

Alexander K. Taylor*, Yicong Huang*, Junheng Hao, Xinyuan Lin, Xiusi Chen, Wei Wang, and Chen Li
40th International Conference on Data Engineering, ICDE 2024 - Dataplat Workshop, pp. Pp. 267–277 [*The first two authors share equal contributions]

D3 Demonstration of Udon: Line-by-line Debugging of User-Defined Functions in Data Workflows



Yicong Huang, Zuozhi Wang, and Chen Li
Companion of the 2024 International Conference on Management of Data, SIGMOD/PODS 2024, Santiago AA, Chile, June 9-15, 2024, pp. Pp. 476–479 [Best Demo Runner-Up Award]

2022

C2 Optimizing Machine Learning Inference Queries with Correlative Proxy Models

Zhihui Yang, Zuozhi Wang, Yicong Huang, Yao Lu, Chen Li, and X. Sean Wang
Proc. VLDB Endow., Vol. 15, No. 10, pp. Pp. 2032–2044

C1 Reshape: Adaptive Result-aware Skew Handling for Exploratory Analysis on Big Data

Avinash Kumar, Sadeem Alsudais, Shengquan Ni, Zuozhi Wang, Yicong Huang, and Chen Li
CoRR, Vol. abs/2208.13143

D2 Demonstration of Collaborative and Interactive Workflow-Based Data Analytics in Texera

Xiaozhen Liu, Zuozhi Wang, Shengquan Ni, Sadeem Alsudais, Yicong Huang, Avinash Kumar, and Chen Li
Proc. VLDB Endow., Vol. 15, No. 12, pp. Pp. 3738–3741

D1 Demonstration of Accelerating Machine Learning Inference Queries with Correlative Proxy Models

Zhihui Yang, Yicong Huang, Zuozhi Wang, Feng Gao, Yao Lu, Chen Li, and X. Sean Wang
Proc. VLDB Endow., Vol. 15, No. 12, pp. Pp. 3734–3737

Interdisciplinary Publications

2025

- 110 DS4ALL: Teaching High-School Students Data Science and AI/ML Using the Texera Workflow Platform as a Service**

Jiadong Bai, Xiaozhen Liu, Anthony Cuturrufo, Alexander Kundu Taylor, Jeehyun Hwang, Mingyu Derek Ma, Xinyuan Lin, Yanqiao Zhu, Yicong Huang, Yunyan Ding, Wei Wang, and Chen Li
Data Science Education K-12: Research to Practice Annual Conference

2024

- 19 Brain image data processing using collaborative data workflows on Texera**

Yunyan Ding, Yicong Huang, Pan Gao, Andy Thai, Atchuth Naveen Chilaparasetti, M Gopi, Xiangmin Xu, and Chen Li
Frontiers in Neural Circuits, Vol. 18, pp. P. 1398884

- 18 Wording Matters: the Effect of Linguistic Characteristics and Political Ideology on Resharing of COVID-19 Vaccine Tweets**

Judith Borghouts, Yicong Huang, Suellen Hopfer, Chen Li, and Gloria Mark
Transactions on Computer-Human Interaction (TOCHI)

- 17 How the experience of California wildfires shape Twitter climate change framings**

Jessie WY Ko, Shengquan Ni, Alexander Taylor, Xiushi Chen, Yicong Huang, Avinash Kumar, Sadeem Alsudais, Zuozhi Wang, Xiaozhen Liu, Wei Wang, et al.
Climatic Change, Vol. 177, No. 1, pp. Pp. 1–21

2023

- 16 Understanding underlying moral values and language use of COVID-19 vaccine attitudes on twitter**

Judith Borghouts, Yicong Huang, Sydney Gibbs, Suellen Hopfer, Chen Li, and Gloria Mark
PNAS nexus, Vol. 2, No. 3, pp. pgad013

- 15 The marketing and perceptions of non-tobacco blunt wraps on Twitter**

Joshua Rhee, Yicong Huang, Sadeem Alsudais, Shengquan Ni, Avinash Kumar, Chen Li, and David Timberlake
Substance Use and Misuse

2022

- 14 Public Opinions toward COVID-19 Vaccine Mandates: A Machine Learning-based Analysis of U.S. Tweets**

Yawen Guo, Jun Zhu, Yicong Huang, Lu He, Changyang He, Chen Li, and Kai Zheng
AMIA 2022, American Medical Informatics Association Annual Symposium, Washington, DC, USA, November 5-9, 2022

- 13 c-myc-mediated upregulation of NAT10 facilitates tumor development via cell cycle regulation in non-small cell lung cancer**

Zimu Wang*, Yicong Huang*, Wanjun Lu, Jiabin Liu, Xinying Li, Suhua Zhu, Hongbing Liu, and Yong Song
Medical Oncology, Vol. 39, No. 10, pp. P. 140 [*The first two authors share equal contributions]

2021

- 12 Why do people oppose mask wearing? A comprehensive analysis of U.S. tweets during the COVID-19 pandemic**

Lu He, Changyang He, Tera L. Reynolds, Qiushi Bai, Yicong Huang, Chen Li, Kai Zheng, and Yunan Chen
J. Am. Medical Informatics Assoc., Vol. 28, No. 7, pp. Pp. 1564–1573

- 11 The social amplification and attenuation of COVID-19 risk perception shaping mask wearing behavior: a longitudinal twitter analysis**

Suellen Hopfer, Emilia J Fields, Yuwen Lu, Ganesh Ramakrishnan, Ted Grover, Qiushi Bai, Yicong Huang, Chen Li, and Gloria Mark
PloS one, Vol. 16, No. 9, pp. e0257428

Research Projects

- 2020 – Pres. *Apache Texera* – Open-source system for cloud-based collaborative data science and AI/ML.
Tech: Scala, TypeScript, Python, Arrow.
Led system design across runtime/compiler/scheduler; built interactive Python UDF debugging and processing engine; maintained hub.texera.io.
- 2019 – 2020 *Cloudberry* – Middleware for large-scale data visualization.
Tech: Scala, JavaScript, AWS.
Built interactive tweet map over multi-terabyte data; integrated COVID and social media data; deployed elastic service across 20+ AWS instances.
- 2019 – 2020 *ML-OPT* – Machine Learning Pipeline Optimization.
Tech: Python.
Explored confidence-aware ML pipeline optimization; improved video/NLP pipelines while preserving high accuracy constraints.
- 2019 *Wildfire* – Wildfire Detection and Visualization with Social Media.
Tech: Python, TypeScript, PostgreSQL.
Led a team of 10 students; built real-time data pipelines from tweets/satellite/fire reports; integrated NLP/CV models for multimodal analysis.
- 2018 *Blockchain* – Blockchain in Fin-tech and Smart Contracts.
Tech: Python, C++.
Reviewed 200+ white papers, developed Ethereum smart contract prototype, and helped organize the California-Shanghai Innovation Dialogue Conference.

Grant Writing Contributions

Under Submission

- 2026 **HCC: ShareWell: From Data Points to Data Owners – Human-Centered Design and Evaluation of Individual-Initiated Sharing of Self-Trackd Health & Wellness Data for Research**
National Science Foundation (NSF), CISE CORE HCC proposal.
Contact PI: Xi Lu (University at Buffalo); Co-PI: Lu He (University of Wisconsin-Milwaukee).
My Role: Unfunded collaborator.
- 2025 **Category II: BRIDGE: A National Center to Democratize Multi-Disciplinary Data Science and AI for Research Labs and Communities**
National Science Foundation (NSF), IDSS proposal.
Contact PI: Chen Li; Co-PIs: Amiri Sani, Shuibing Chen, Turney, and Wei Wang.
My Role: Unfunded collaborator.

Granted

- 2024 **dkNET Coordinating Unit: Harnessing the Power of AI and Data Science for Collaborative Discovery and Sharing in the DK Community**
National Institutes of Health (NIH), National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK).
Contact PI: Jeffrey S. Grethe; Co-PIs: Shuibing Chen, Chen Li, Wei Wang.
Period: 2024 – 2029; Funding: total \$10M for 5 years.
My Role: Contributed to Aim 3 (Computational Core): unified ML models and workflows.
- 2022 **Collaborative Research: Collaborative Machine-Learning-Centric Data Analytics at Scale**
National Science Foundation (NSF).
Contact PI: Chen Li; Co-PI: Suellen Hopfer.
Period: 2021 – 2024; Funding: \$913,992.
My Role: Contributed to Aim 2: supporting debugging of external UDFs during execution.
- 2020 **RAPID: Leveraging Twitter Data for Real-time Public Health Responses to Coronavirus**
National Science Foundation (NSF).
Contact PI: Gloria Mark; Co-PIs: Chen Li, Suellen Hopfer.
Period: 2020; Funding: \$180,000.
My Role: Collected data and contributed to writing preliminary results.

Service

Conference Officer

2027 Web Chair, SIGMOD

Reviewer

2027 Program Committee, SIGMOD

2026 Program Committee, SIGMOD

2024 External Reviewer, IEEE Transactions on Knowledge and Data Engineering (TKDE)

2024 Program Committee, SIGMOD – Artifact Review and Reproducibility (ARI)

Student Volunteer

2023 ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD)

2023 IEEE International Conference on Data Engineering (ICDE)

2019 International Conference on Very Large Data Bases (VLDB)

2018 California-Shanghai Innovation Dialogue Conference

Teaching

Workshop Instructor

2024 Cerritos College, Norwalk, CA, United States

Fall'24 *Workshop of Data Science for Everyone 2024* ([Workshop Homepage](#)); two-day no-code data science/ML workshop; 50+ community college students and faculty.

Associate Instructor (Lecturer)

2024 UC Irvine, CA, United States

Spring'24 *ICS 80: Data Science and AI/ML Using Workflows* ([Syllabus](#)); designed and taught new course for non-CS students; 42 undergraduates enrolled.

Teaching Assistant

2018 – 2022 UC Irvine, CA, United States

Winter'22 *CS 222/122C: Principles of Data Management*

Fall'21 *CS 122B: Projects in Databases and Web Applications*

Spring'21 *CS 122B: Projects in Databases and Web Applications*

Winter'21 *ICS 51: Introduction to Computer Organization*

Fall'20 *CS 222/122C: Principles of Data Management*

Spring'20 *CS 122B: Projects in Databases and Web Applications*

Winter'20 *CS 222/122C: Principles of Data Management*

Fall'19 *CS 222/122C: Principles of Data Management*

Spring'19 *CS 122B: Projects in Databases and Web Applications*

Winter'19 *CS 122B: Projects in Databases and Web Applications*

Fall'18 *CS 141: Concepts of Programming Languages I*

Spring'18 *CS 122B: Projects in Databases and Web Applications*

Mentor

Winter'18 Dreams for Schools APPJAM+, Yorba Linda High School, Yorba Linda, CA, United States

Tutor

2016 – 2018 UC Irvine, CA, United States

Winter'18 *ICS 46: Data Structure Implementation and Analysis*

Fall'17 *ICS 45J: Programming in Java as a Second Language*

Spring'17 *ICS 33: Intermediate Programming*

Winter'17 ICS 32: Programming with Software Libraries

Fall'16 ICS 31: Introduction to Programming

Mentorship

Ph.D. Student Mentorship

2025 – 2026	Jiadong Bai	–	D4, I10
2023 – 2024	Raj Mohanty	–	–
	Shagoto Rahman Shrestho	–	–
2022 – 2023	Xinyuan Lin	–	C5, C4, W1, D4, I10
	Yunyan Ding	–	C4, I9, I10

M.S. Research Mentorship

2022 – 2023	Aditya Verma	–	B2
	Sreetej Reddy	–	B2
	Dhruv Raipure	–	B2
	Jiaxi Chen	–	B2
2019 – 2020	Yang Cao	University of California, Irvine, MS	–

Undergraduate Research Mentorship

2023 – 2024	Kevin Wu	–	B3
2022 – 2023	Chengxi Li	–	–
	Ethan Wong	–	–
	Tianyun Yuan	University of Illinois Urbana-Champaign, MSCS	–
	Tony Liu	University of California, San Diego, MSCS	–
2021 – 2022	Zhen Guan	University of California, San Diego, MS	–
	Jiashu Zhang	Hong Kong Polytechnic University, PhD	–
	Yinan Zhou	University of California, Irvine, PhD	–
	Andrew Li	–	–
	Eric Peng	–	–
	Jiyang Wu	–	–
	Zeyu Li	–	–
2020 – 2021	Chen He	Carnegie Mellon University, MS	–
	Bihao Xu	University of Chicago, MS	–
	Conghuai Tan	–	–
	Make Tao	–	–
	Mingshuo Liu	–	–
	Qifan Yu	–	–

2019 – 2020	Dayue Bai	University of Illinois Urbana-Champaign, MS	–
	Yinan Zhou	University of California, Irvine, MS	–
	Shiqi Wu	University of California, Berkeley, MS	–
	Christine Xinrong Huang	Carnegie Mellon University, MS	–
	Tianran Liu	University of Washington, MS	–
	Yutong Wang	University of California, Davis, PhD	–
	Tingxuan Gu	Carnegie Mellon University, MS	–
	Yichi Zhang	New York University, MS	–
	Xinyue Han	University of California, Los Angeles, MS	–
	Qiaonan Huang (Hugo)	Brown, MS	–
	Yuan Fu	Carnegie Mellon University, SE-SV	–
	Yuqi Huai	University of California, Irvine, PhD	–
	Quanzhen Du	University of California, San Diego, MS	–
	Shiling (Scarlett) Zhang	Cornell, MS	–
	Zeyad Kelani	–	–

Blogs

2024

B4 Suggesting Python Code Type Annotations with LLM – A Language Server Enhancement

Minchong Wu and [Yicong Huang](#)

Texera Blog

B3 Adding R UDF to Texera: The Journey

Kevin Wu and [Yicong Huang](#)

Texera Blog

2023

B2 Enhancing the UDF Editor by Adding Language Server Support

Aditya Verma, Dhruv Raipure, Jiayi Chen, Sreetej Reddy, and [Yicong Huang](#)

Texera Blog

B1 Using Texera to Perform Single-cell RNA Sequencing Analysis with R Language

[Yicong Huang](#)

Texera Blog

Invited Talks

2025 UDF-Centric Dataflow Systems for Supporting User-Defined Functions in Collaborative Data Science, AI, and ML

Final defense, UC Irvine

Building a Collaborative and Interactive Data System to Broaden Access to Data Science, AI, and ML

University at Buffalo

Lehigh University

UMass Amherst

Purdue University

UMass Boston

Amazon Web Services

Arizona State University

Santa Clara University

2024 Texera and Interactive Debugging in Distributed Data Processing Systems

Invited talk, Databricks, Mountain View, CA

Texera: A System for Collaborative and Interactive Data Analytics Using Workflows

Paper presentation, VLDB 2024, Guangzhou, China

Udon: Efficient Debugging of User-Defined Functions in Big Data Systems

Paper presentation, SIGMOD 2024, Santiago, Chile

The Journey to Build Texera

Invited talk, ISG Reunion, Irvine, CA

2023 Python User-defined Functions on Workflows

Guest lecture, DS4ALL Summer Program, UC Irvine

2022 Data Challenges in Streaming Systems

VISA Research AI Research Team, Palo Alto, CA

Computer Skills

Proficient	Python, Java/Scala, C/C++, JavaScript/TypeScript, Golang, Lisp, Prolog, MySQL, PostgreSQL
Intermediate	HTML, CSS, SQL, R, Redis
Concepts	MapReduce, Object-Oriented Design, Functional Programming, Logical Programming, Model-View-Control, Multithreading
Frameworks	Arrow, Hadoop, Spark, Flink, Kudu, Protobuf, Spring, Angular 2+, Django, Flask, Express, ReactJS, Elastic-search
Services	AWS, GCP, GitHub, Docker